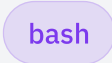


Module 2 Cheat Sheet - Introduction to Linux Commands

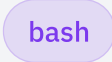
Getting information

Return your user name:

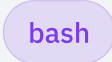
`whoami` 

Return your user and group id:

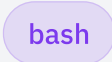
`id` 

Return operating system name, username, and other info:

`uname -a` 

Display reference manual for a command:

`man top` 

List available  pages, including a brief description for each command:

bash

```
man -k .
```

Run

Get help on any command (for eg: `curl`):

bash

```
curl --help
```

Run

This provides a brief overview of the `curl` command's usage and options.

Return the current date and time:

bash

```
date
```

Run

Navigating and working with directories

List files and directories by date, newest to last:

bash

```
ls -lrt
```

Run

Find files in directory tree that end in `.sh`:

bash

```
find -name "*.sh"
```

Run

Return path to present working directory:

bash

```
pwd
```

Run

Make a new directory:

bash

```
mkdir new_folder
```

Run

Change the current directory:

Up one level:

bash

```
cd ../
```

Run

To home:

bash

```
cd ~
```

Run

or

bash

```
cd
```

Run

To some other directory: `cd path_to_directory`

Remove directory verbosely:

bash

```
rmdir temp_directory -v
```

Run

Monitoring system performance and status

List selection of/all running processes and their PIDs:

ps



ps -e



Display resource usage:

top



List mounted file systems and usage:

df



Creating, copying, moving, and deleting files:

Create an empty file or update existing file's timestamp:

touch a_new_file.txt



Copy a file:

bash

```
cp file.txt new_path/new_name.txt
```

Run

Change file name or path:

bash

```
mv this_file.txt that_path/that_file.txt
```

Run

Remove a file verbosely:

bash

```
rm this_old_file.txt -v
```

Run

Working with file permissions

Change/modify file permissions to 'execute' for all users:

bash

```
chmod +x my_script.sh
```

Run

Change/modify file permissions to 'execute' only for you, the current user:

bash

```
chmod u+x my_file.txt
```

Run

Remove 'read' permissions from group and other users:

bash

```
chmod go-r my_file.txt
```

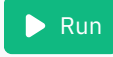
Run

Displaying file and string contents

Display file contents:

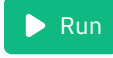
```
cat my_shell_script.sh
```



Display file contents page-by-page:

```
more ReadMe.txt
```



Display first 10 lines of file:

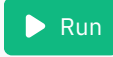
```
head -10 data_table.csv
```



Display last 10 lines of file:

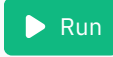
```
tail -10 data_table.csv
```



Display string or variable value:

```
echo "I am not a robot"  
echo "I am $USERNAME"
```



Basic text wrangling

Sorting lines and dropping duplicates:

bash

```
sort text_file.txt
```

Run

In reverse order:

bash

```
sort -r text_file.txt
```

Run

Drop consecutive duplicated lines and display result:

bash

```
uniq list_with_duplicated_lines.txt
```

Run

Displaying basic stats:

Display the count of lines, words, or characters in a file:

Lines:

bash

```
wc -l table_of_data.csv
```

Run

Words:

bash

```
wc -w my_essay.txt
```

Run

Characters:

bash

```
wc -m some_document.txt
```

Run

Extracting lines of text containing a pattern:

Some frequently used options for `grep`:

Option	Description
-n	Print line numbers along with matching lines
-c	Get the count of matching lines
-i	Ignore the case of the text while matching
-v	Print all lines which do not contain the pattern
-w	Match only if the pattern matches whole words

Extract lines containing the word "hello", case insensitive and whole words only:

bash

```
grep -iw hello a_bunch_of_hellos.txt
```

Run

Extract lines containing the pattern "hello" from all files in the current directory ending in `.txt`:

bash

```
grep -l hello *.txt
```

Run

Merge two or more files line-by-line, aligned as columns:

Suppose you have three files containing the first and last names of your customers, plus their phone numbers.

Use `paste` to align file contents into a Tab-delimited table, one row for each customer:

bash

```
paste first_name.txt last_name.txt phone_number.txt
```

Run

Use a comma as a delimiter instead of the default Tab delimiter:

bash

```
paste -d "," first_name.txt last_name.txt phone_number.txt
```

Run

Use the `cut` command to extract a column from a table-like file:

Suppose you have a text file whos rows consist of first and last names of customers, delimited by a comma.

Extract first names, line-by-line:

bash

```
cut -d "," -f 1 names.csv
```

Run

Extract the second to fifth characters (bytes) from each line of a file:

bash

```
cut -b 2-5 my_text_file.txt
```

Run

Extract the characters (bytes) from each line of a file, starting from the 10th byte to the end of the line:

bash

```
cut -b 10- my_text_file.txt
```

Run

Compression and archiving

Archive a set of files:

 bash 

```
tar -czvf my_archive.tar.gz file1 file2 file3
```

 Run

Compress a set of files:

 bash 

```
zip my_zipped_files.zip file1 file2  
zip my_zipped_folders.zip directory1 directory2
```

 Run

Extract files from a compressed zip archive:

 bash 

```
unzip my_zipped_file.zip  
unzip my_zipped_file.zip -d extract_to_this_directory
```

 Run

Working with networking commands

Print hostname:

 bash 

```
hostname
```

 Run

Send packets to URL and print response:

 bash 

```
ping www.google.com
```

 Run

Display or configure system network interfaces:

bash

```
ip a
```

Run

Display contents of file at a URL:

bash

```
curl <url>
```

Run

Download file from a URL:

bash

```
wget <url>
```

Run

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▶ Run

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